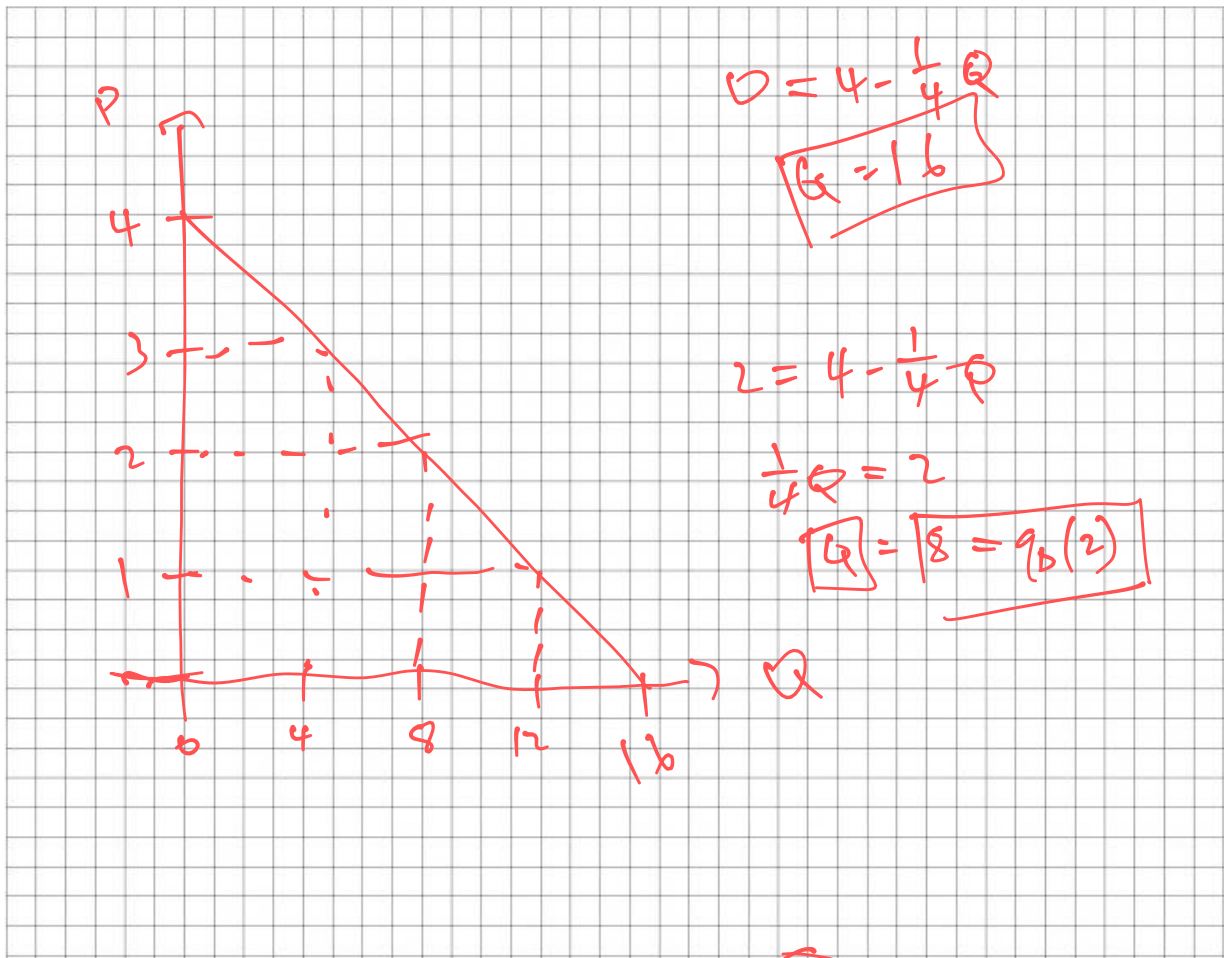


Homework 3 Demo

Name Taylor

This is a set of practice questions similar to HW 3. I'm very happy to answer questions.

Q1. (Tutorial 2.1) The demand curve for carrots is $P = 4 - \frac{1}{4}Q$. Use a graph to plot this demand curve, and find the quantity demanded at a price of 1, 2, and 3.



Quantity demanded at a price of 1 Knuts 12

Quantity demanded at a price of 2 Knuts 8

Quantity demanded at a price of 3 Knuts 4

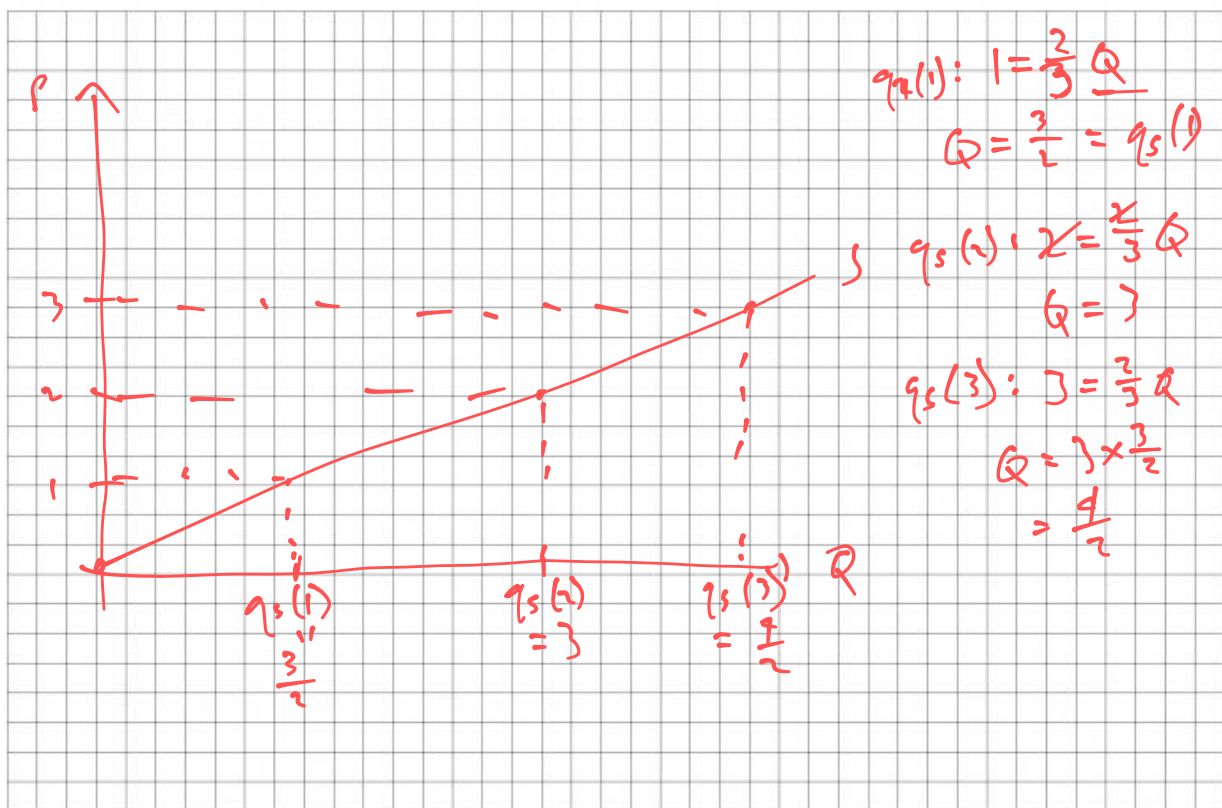
Q2. (Tutorial 2.5) Use the midpoint method to find the elasticity of demand when the price changes from 1 to 2.

$$\epsilon_D = \frac{\% \Delta Q_D}{\% \Delta P} =$$

$$= \frac{\frac{-4}{16}}{\frac{2-1}{\frac{3}{2}}} = \frac{-\frac{1}{4}}{1 - \frac{2}{3}} = \frac{-\frac{1}{4}}{\frac{1}{3}} = -\frac{3}{4}$$

Elasticity of Demand (1 to 2) $-\frac{3}{4}$

Q3. (Tutorial 2.2) The supply curve for carrots is $P = \frac{2}{3} Q$. Use a graph to plot this supply curve, and find the quantity demanded at a price of 1, 2, and 3.



Quantity supplied at 1 $\frac{3}{2}$

Quantity supplied at 2 3

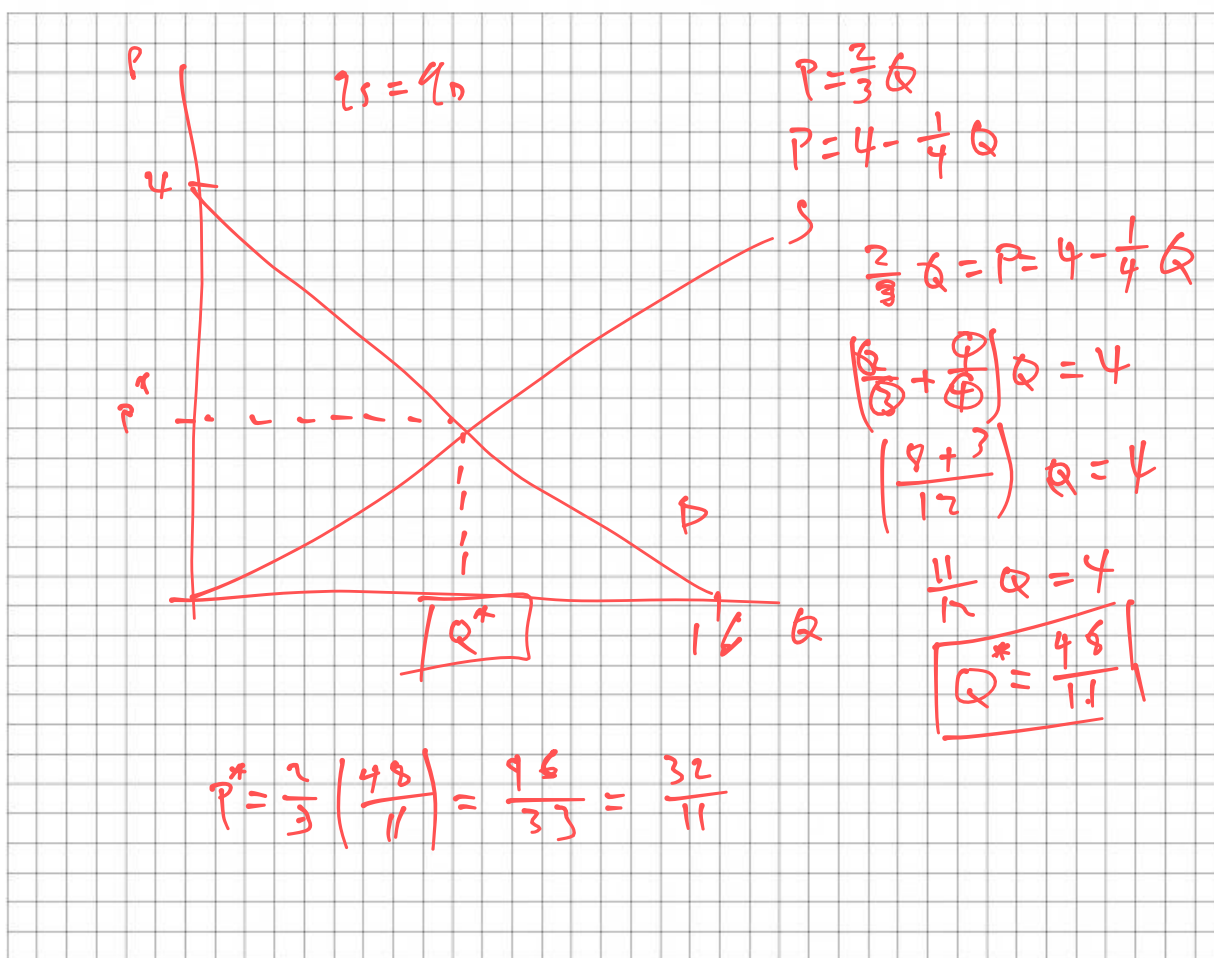
Quantity supplied at 3 $\frac{9}{2}$

Q4. (Tutorial 2.5) Use the midpoint method to find the elasticity of supply when the price changes from 1 to 2.

$$\epsilon_s = \frac{\% \Delta Q_s}{\% \Delta P} = \frac{\frac{q_2 - q_1}{\frac{q_2 + q_1}{2}}}{\frac{2 - 1}{\frac{1 + 2}{2}}} = \frac{3}{2} \times \frac{3 - \frac{3}{2}}{\frac{3 + \frac{3}{2}}{2}} = \frac{3}{2} \times \frac{\frac{6 - 3}{2}}{\frac{1 + 3}{2}} = \frac{3}{2} \times \frac{\frac{3}{2}}{2} = \frac{3}{2} \times \frac{3}{4} = \frac{9}{8} = 1.125$$

Elasticity of Supply (1 to 2) 1.125

Q5. (Tutorial 2.3) Use the supply and demand curves from Q1 and Q2 to find the equilibrium price and quantity in the carrot market.

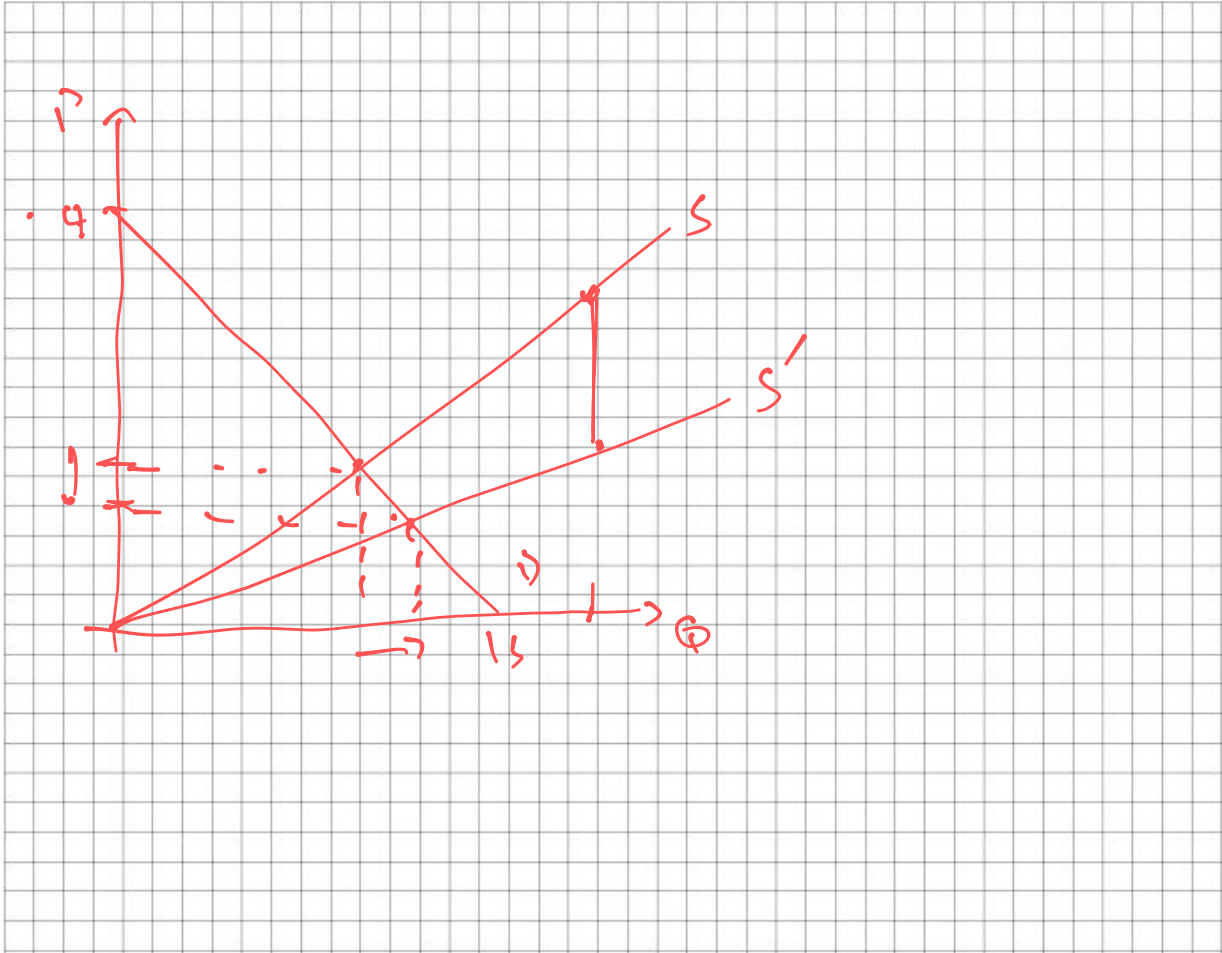


Equilibrium Quantity $\frac{48}{35}$

Equilibrium Price $\frac{32}{35}$

$$P = 4 - \frac{1}{4} \left(\frac{48}{35}\right) = 4 - \frac{12}{35} = \frac{140 - 12}{35} = \frac{128}{35}$$

Q6. (Tutorial 2.4) Use a graph to discuss what happens if carrot growers suddenly discover a miracle fertilizer that significantly increases harvests. The new supply curve is $P = 1/3 Q$. Using a graph, illustrate what happens to the market, equilibrium price, and quantity?



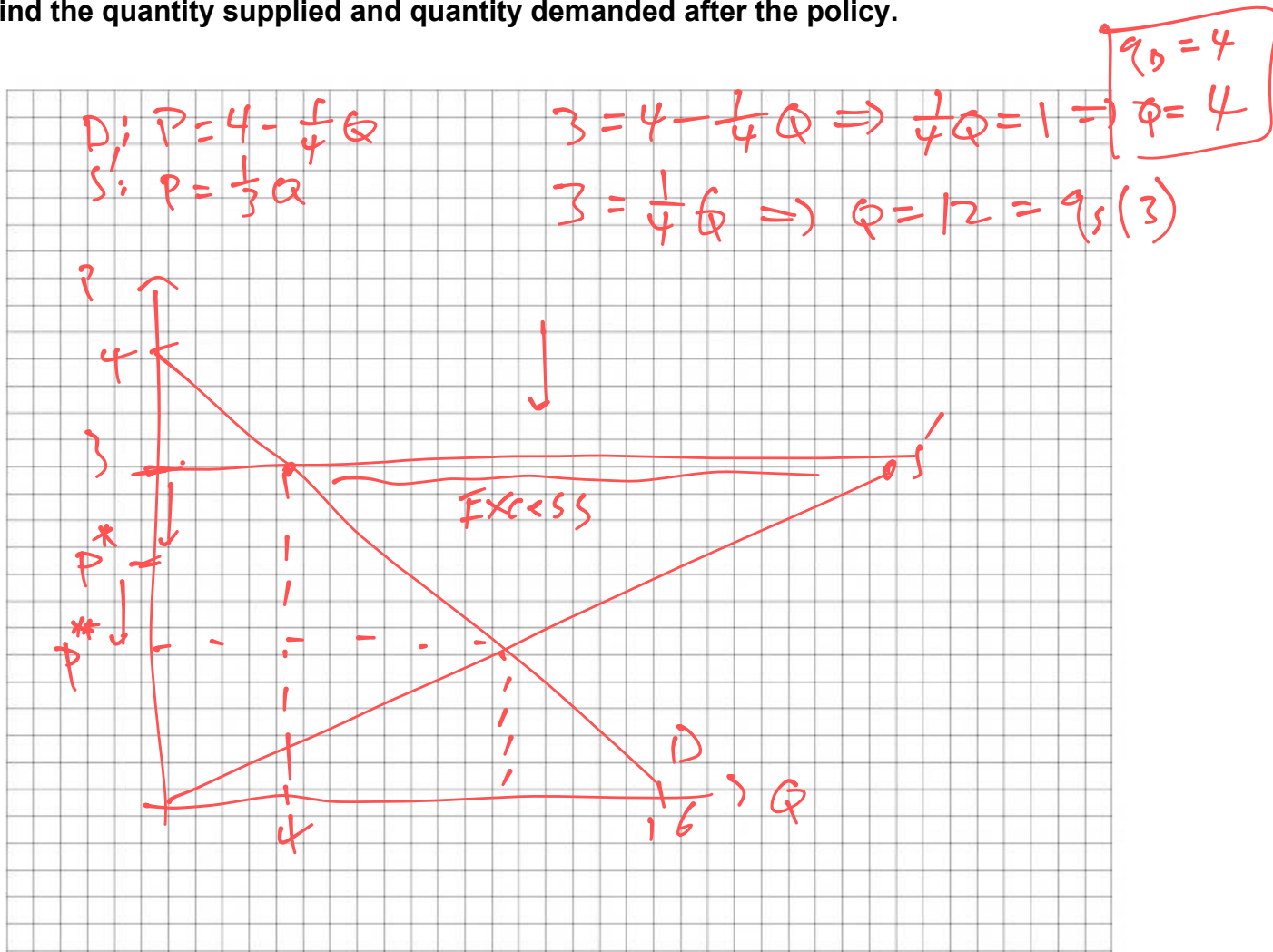
Equilibrium Price (increase, decrease, no change, indeterminate)

Decrease

Equilibrium Quantity (increase, decrease, no change, indeterminate)

Increase

Q7. (Tutorial 2.6) After the fertilizer discovery, the government grew concerned that the price of carrots had become too low and was bad for farmers, and decided to institute a price floor of 3. Use a graph of the original demand curve and the new supply curve and the policy to illustrate what effect this policy, and find the quantity supplied and quantity demanded after the policy.



Quantity Supplied 12

Quantity Demanded 4

Excess, Market Clearing, or Shortage EXCESS